To: Ms. Letrari Liu • Section Managing Editor
The Editorial Office & Editor-in-Chief • *International Journal of Molecular Sciences* (IJMS), MDPI
St. Alban-Anlage 66, 4052 Basel, Switzerland

SUBJECT: Invited Submission — "General Scaling Laws and Thermodynamically Consistent Reduced Stochastic Operators for Macromolecular Protein Dynamics" (Attn: Ms. Letrari Liu)

Dear Ms. Letrari Liu and Editors of IJMS,

We are pleased to submit our original research manuscript entitled **"General Scaling Laws and Thermodynamically Consistent Reduced Stochastic Operators for Macromolecular Protein Dynamics"** for consideration as an invited Research Article in the *International Journal of Molecular Sciences* (IJMS), following the editorial invitation from **Ms. Letrari Liu**.

Coarse-graining macromolecular dynamics from atomistic molecular dynamics (MD) while maintaining microscopic reversibility, non-Markovian memory, and computational scalability is a fundamental challenge in molecular biophysics. Standard methods frequently fail near transition barriers (Kramers turnover) or violate detailed balance. In this work, we present a unified theoretical and computational framework evaluated across seven molecular benchmark systems spanning model peptides (dialanine, trialanine) to five structurally diverse proteins (d = 2 to 150 conformational dimensions: Chignolin, Trp-cage, Villin HP35, GB1, and Human Ubiquitin). Our primary findings include:

1. **Sub-linear operator rank scaling:** The effective dynamical rank of the diffusion operator scales sub-linearly with chain length ($r_{90}(d) = 0.745 \cdot d^{0.887}$, 95% bootstrap CI [0.830, 0.950], rejecting linear growth at $P(\alpha < 1.0) = 99.95\%$). An uncoupled harmonic null model recovers linear scaling ($\alpha_{null} \approx 1.01 \pm 0.02$), demonstrating that compression is an emergent macroscopic property of cooperative interactions.
2. **Dynamic locality and tertiary contact gating:** Off-diagonal diffusion couplings decay as a power law along the sequence ($|i - j|^{-1.34}$), while native tertiary contacts gate strong off-diagonal coupling with a 2.1- to 3.9-fold enrichment over non-contact pairs.
3. **Asymptotic linear-time estimation:** Exploiting contact-gated locality yields an $O(d^{0.99})$ sparse estimator that accelerates generator estimation by 28-fold at d = 2,400 with under 1.8% spectral error.
4. **Invariant interior mobility floor:** Core residues converge to an invariant mobility plateau ($\bar{D} = 0.0496 \pm 0.0016$ rad^2/ps, cross-protein CV 8.0%) that correlates significantly with experimental 1.8 Å X-ray crystallographic Debye-Waller B-factors ($r_s = 0.272$, $p = 0.019$).
5. **Thermodynamic consistency:** Embedding dynamics into an extended Mori-Zwanzig state space (z, v, w) with structure-preserving GENERIC drift eliminates 90.8% of artificial entropy production and captures atomistic folding barrier-crossing times within the 95% bootstrap confidence interval.

The manuscript is accompanied by a complete 75-page Supporting Information Appendix containing all mathematical derivations, 61 high-resolution figures, audit matrices, and cryptographic checksums. All code, models, and simulation pipelines are open-source and immediately accessible.

This work is original, has not been published previously, and is not currently under consideration elsewhere. Both authors have approved the submission. We look forward to working with Ms. Letrari Liu and the IJMS editorial team.

Thank you very much for your kind invitation and editorial evaluation.

Sincerely,

Mehardeep Singh
Sri Chaitanya Educational Institutions
Chandigarh, India

Gurarpit Singh
Vivekananda Institute of Professional Studies (VIPS)
Guru Gobind Singh Indraprastha University (GGSIPU)
Delhi, India